

Piotr Antoniak

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WORK EXPERIENCE

Data Scientist

May 2024 - ongoing

UBS; Associate Director

Zurich, Switzerland

- Provided various analysis and reconciliations for a for the first joint CS-UBS sustainability report
- Tested series of overseas developed LLM based workflows
- Developed POC LLM based programs to automate pdf filling and corroboration of client documents
- Automated manual, error prone excel routines
- Assisted colleagues with ad-hoc request related to reconciliation and supporting already delivered projects

Data Scientist

July 2023 - May 2024

Deloitte; Audit & Assurance, Assistant: references

Zurich, Switzerland

- Built a pipeline for reporting ECL and Exposure related disclosures (including so-called movement table) for a large Swiss bank
- Improved ETL system extracting data from client's SAP and trading system from excel based to combination of SQL and python. This resulted in decrease of time needed to update a dashboard from 4 hours to just a few minutes
- Built system to extract and analyze the data from scanned documents submitted for loan applications
- Improved already existing SQL code and Power Bi dashboards

Analyst

April 2022 - June 2023

European Central Bank; MacroPrudential & Financial Stability, Analyst; references

Franfurt am Main, Germany

- Developed an end to end, top-down, reverse stress testing model that allows to simulate thousands of scenarios and test multiple parallel constrains (i.e. solvency, leverage, liquidity) of Eurozone banks. It utilizes exposures to interest rates, sovereign bonds, economy on industrial levels and unconventional monetary policy. Model's results are included in bi-yearly ECB's MacroPrudential Report and presented in front of all EuroArea Central Banks Financial Stability/MacroPrudential representatives
- Fulfilled ad-hoc data requests from various, highly granular databases using SQL
- Management of data pipelines utilized by members of my teams

Data Scientist

October 2021 - April 2022

BMW; Financial Services, Intern; references

Munich, Germany

- Developed proof-of-concept of a search and retrieval system for internal data drive and confluence page
- Participated in development and deployment of a fraud detection model covering an entire continent - over 80 000 new customers yearly
- Developed ML model and deployed it as a proof-of-concept for Customer Feedback division. It was trained to classify both sentiment and type of incoming feedback and was an improvement in terms of capabilities and accuracy over previously used rule based model
- Built a few dashboards/simple pages using streamlit/flask serving as a GUI for ML models

EDUCATION

Pompeu Fabra University and Universitat Autònoma de Barcelona

September 2020 - July 2021

Master's Degree in Macroeconomic Policy and Financial Markets

Barcelona, Spain

Thesis: Predictive power in online investing chatter

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Aberystwyth University

September 2017 - June 2020

Bachelor's Degree in Economics

Aberystwyth, United Kingdom

Thesis: Evaluation of the post-crisis monetary policy in the USA using Taylor rules

First class

PERSONAL PROJECTS

Tribal-Wars Snipes | *python, sql, streamlit*

- Website that allowed my teammates to optimize defense planning and execution in a strategy, web based game.
- Despite being distributed to limited amount of players, I estimate it was able to save 200-300 hours of manual work on a weekly basis.

Open Domain Question Answering; working demo | *python, torch, streamlit, GPT*

- A tool that can answer questions using documents or videos given to it

Simple chatbot from scratch | *python, torch, flask*

- A simple chat bot that uses predefined question-answer pairs
- Its simplicity is based on the fact that extending usage is just a matter of updating a .csv file

Predictive power in online investing chatter | *python, tensorflow, torch, pandas, data streams*

- My Master Thesis
- I investigated millions of comments on investing/trading forums in order to extract sentiment and check whether users retail traders are so unskilled as claimed by various studies

GPT-2 twitter bot mimicking Central Banks speeches | *python, torch, scrapping, twitter API, Docker, AWS*

- A Twitter Bot based on GPT-2 further pre-trained on speeches from major Central Banks

OTHER

- Computer skills: Python, MATLAB, SQL, Alteryx, Spark/PySpark
- Languages: Polish (native), English
- T Maelgwyn Davies Prize - Best Economics Dissertation (Aberystwyth, June 2020)